

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

Angela Zavaleta Bernuy



Surjective

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Example.

- Is $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ surjective? **X**

$b \in \mathbb{R}$, let $b = -4$ $f(a) = -4 = x^2 \rightarrow a = \sqrt{-4}$
 $\rightarrow a \notin \mathbb{R}$.

- Is $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = -3x^2 + 7$ surjective? **X**

$b = 20$ $f(a) = 20$
 $-3a^2 + 7 = 20 \rightarrow -3a^2 = 13$
 $a^2 = -13/3$

Bijjective

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Example.

Is $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3 + 1$ bijective?

injective

Let $x_1, x_2 \in \mathbb{R}$

$$x_1 < x_2 \rightarrow f(x_1) < f(x_2)$$

$$x_1^3 + 1 < x_2^3 + 1$$

$$x_1^3 < x_2^3$$

$$x_1 < x_2$$

$\therefore f(x)$
is strictly
inc.

$\rightarrow f(x)$ is
injective

surjective

$$b \in \mathbb{R}, \exists a \text{ s.t. } f(a) = b$$

$$a^3 + 1 = b$$

$$a^3 = b - 1$$

$$a = \sqrt[3]{b-1} \in \mathbb{R}$$

$\rightarrow f(x)$ is surjective.
 $\square \rightarrow \checkmark$ bijective.

Last Lecture:

- Injective functions: $a \neq b \rightarrow f(a) \neq f(b)$
- Increasing: $a < b \rightarrow f(a) \leq f(b)$
- Strictly increasing: $a < b \rightarrow f(a) < f(b)$
- Decreasing: $a < b \rightarrow f(a) \geq f(b)$
- Strictly decreasing: $a < b \rightarrow f(a) > f(b)$
- **Theorem:** Every strictly increasing function is injective.
- Surjective function: for $f : A \rightarrow B$, we have $\forall(b \in B)\exists(a \in A)(f(a) = b)$
- Bijective: injective and surjective

Necessary condition

If $f : A \rightarrow B$ is bijective, then:

- Range of f = Codomain B

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- Range of f = Codomain B

- Domain and Codomain have the same size: $|A| = |B|$.

↳ a perfect pairing between elements of A and B .

- The cardinality of the preimage of each $b \in B$ is one.

Every output comes from exactly one input.

Composition

The composition of functions $f : \underline{A} \rightarrow \underline{B}$ and $g : \underline{B} \rightarrow \underline{C}$ is defined by

$$\begin{aligned} \underline{g} \circ f : \quad A &\longrightarrow C \\ x &\longmapsto g(f(x)) \end{aligned}$$

Example

- $f : \mathbb{R} \rightarrow \underline{\mathbb{R}}$ given by $f(x) = x^2 + 1$
- $g : \underline{\mathbb{R}} \rightarrow \mathbb{R}$ given by $g(x) = x + 2$

What is $g \circ f$?

$$\begin{aligned} = g(f(x)) &= (x^2 + 1) + 2 \\ &= x^2 + 3 \end{aligned}$$

Inverse function

Definition

Suppose $f : A \rightarrow B$ is bijjective. The inverse of f is a function $g : B \rightarrow A$ such that $(g \circ f)(x) = \underline{x}$ for all $x \in A$.

- We usually denote the inverse of f by f^{-1} .

Warning

- $f^{-1}(x) \neq \frac{1}{f(x)}$
- If f is not bijective, then its inverse is not defined.

Example

Find the inverses of the following:

- $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 1$.

linear $\rightarrow$ injective
 $\rightarrow$ range $\mathbb{R} \rightarrow$ surjective } bijection

$$y = 2x + 1$$

$$y - 1 = 2x$$

$$x = \frac{y-1}{2}$$

$$f^{-1}(x) = \frac{x-1}{2}$$

- $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^5 + 1$.

linear continuous
 $\rightarrow$ range $\mathbb{R}$

$$y = x^5 + 1$$

$$y - 1 = x^5$$

$$x = \sqrt[5]{y-1}$$

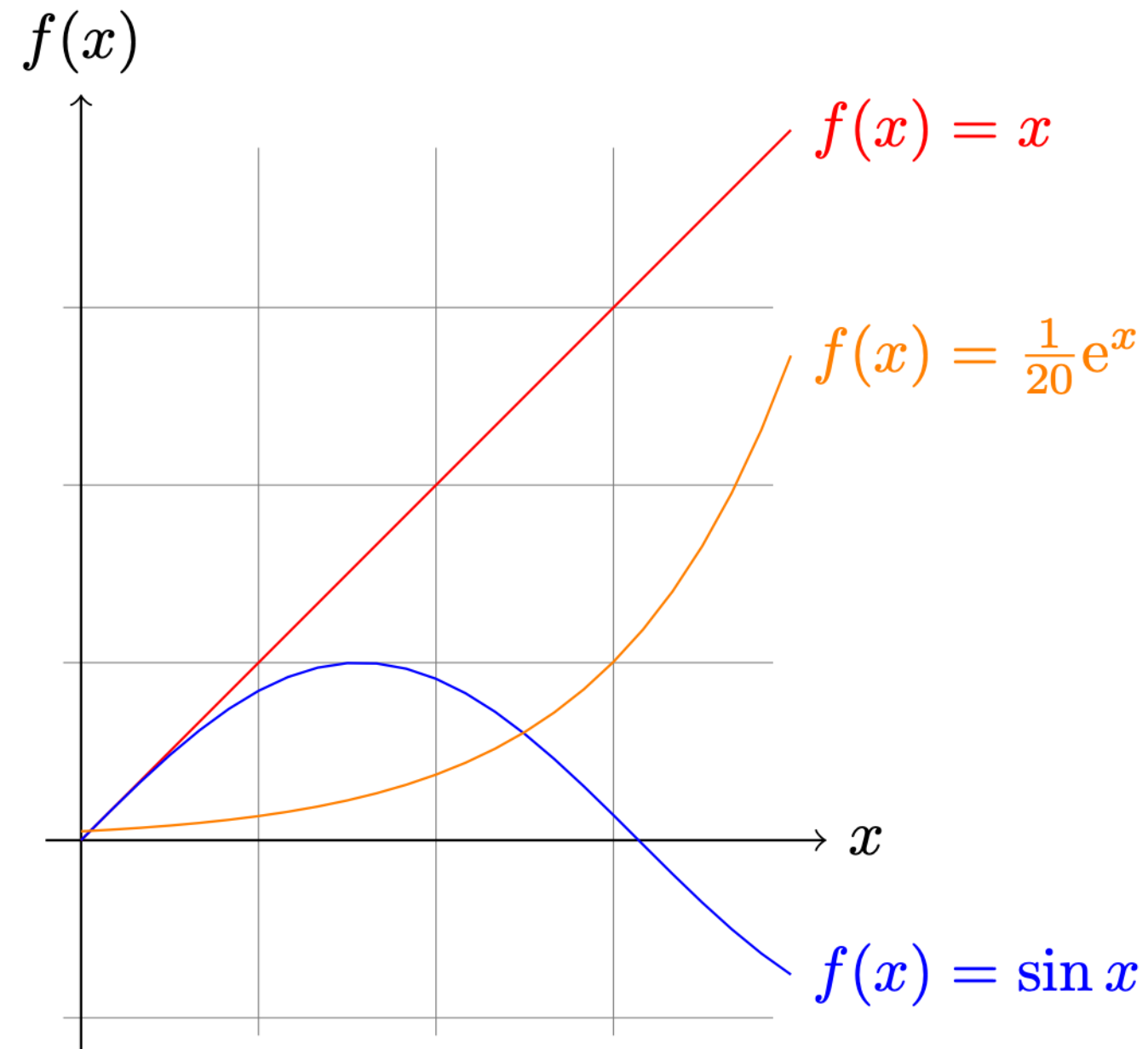
$$f^{-1}(x) = \sqrt[5]{x-1}$$

The Graph of a Function

Definition

The graph of a function $f : A \rightarrow B$ is a set of ordered pairs $\{(a, f(a)) \mid a \in A\}$.

- When $A = B = \mathbb{R}$, we can draw the graph:



Some important functions

- Floor function: $f : \mathbb{R} \rightarrow \mathbb{Z}$ is defined as

$$f(x) = \lfloor x \rfloor = \text{largest integer } \leq x$$

Example: $\lfloor 3.1 \rfloor = 3$, $\lfloor -1.1 \rfloor = -2$

- Ceiling function: $f : \mathbb{R} \rightarrow \mathbb{Z}$ is defined as

$$f(x) = \lceil x \rceil = \text{smallest integer } \geq x$$

Example: $\lceil 3.1 \rceil = 4$, $\lceil -1.1 \rceil = -1$

- Factorial: $f : \mathbb{N} \rightarrow \mathbb{Z}^+$ defined as

$$\rightarrow 0! = 1$$

$$f(n) = \underline{n!} = \underline{n} \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$$

Example: $5! = 120$, $13! = 6227020800$

$$\hookrightarrow 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Properties

- $\lfloor x \rfloor = n \iff n \leq x < n + 1 \rightarrow x \in [n, n+1)$
- $\lceil x \rceil = n \iff n - 1 < x \leq n \rightarrow x \in (n-1, n]$
- For any real number x :

$$x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$$

- For any integer $n \in \mathbb{Z}$:

$$\begin{aligned} & \lfloor 3.1 + 4 \rfloor \\ &= \lfloor 3.1 \rfloor + 4 \end{aligned}$$

$$\lfloor x + n \rfloor = \lfloor x \rfloor + n, \quad \lceil x + n \rceil = \lceil x \rceil + n$$

- For any real $x \in \mathbb{R}$:

$$\lfloor -x \rfloor = -\lceil x \rceil, \quad \lceil -x \rceil = -\lfloor x \rfloor$$

Example

$$\rightarrow \lfloor 2n + 2r \rfloor = 2n + \lfloor 2r \rfloor$$

$$\rightarrow \lfloor n + r + \frac{1}{2} \rfloor = n + \lfloor r + \frac{1}{2} \rfloor$$

Prove that $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$ for all $x \in \mathbb{R}$.

$$\underbrace{n = \lfloor x \rfloor}, \quad x = n + \underline{r}, \quad \underline{0 \leq r < 1}, \quad n \in \mathbb{Z}$$

$$\rightarrow \cancel{2n} + \lfloor 2r \rfloor = \cancel{n} + \cancel{n} + \lfloor r + \frac{1}{2} \rfloor$$

$$\textcircled{\lfloor 2r \rfloor = \lfloor r + \frac{1}{2} \rfloor}$$

$$\text{Case 1: } 0 \leq r < \frac{1}{2} \rightarrow \frac{1}{2} \leq \underline{r + \frac{1}{2}} < 1$$

$$0 \leq 2r < 1$$

$$\lfloor 2r \rfloor = 0$$

$$\lfloor \underline{r + \frac{1}{2}} \rfloor = 0$$



$$\text{Case 2: } \frac{1}{2} \leq r < 1$$

$$\underline{1 \leq 2r < 2}$$

$$\lfloor 2r \rfloor = 1$$

$$\checkmark \lfloor r + \frac{1}{2} \rfloor = 1$$



□

Example

Let $n = \lfloor x \rfloor$, $\boxed{n \leq x < n+1}$, $n \in \mathbb{Z}$

Prove that for all $x \in \mathbb{R}$ and $\underline{m \in \mathbb{Z}}$

Case 1: $m > 0$

$$mn \leq \underline{mx} < mn + m$$

$$\lfloor mn \rfloor = mn$$

$$mx < \underline{mn + m}$$

$$\lfloor mx \rfloor \leq mn + m - 1$$

$$mn \leq \lfloor mx \rfloor \leq mn + m - 1$$

$$n \leq \frac{\lfloor mx \rfloor}{m} \leq n + \frac{m-1}{m} < n+1$$

$$\frac{\lfloor mx \rfloor}{m} \in [n, n+1) \rightarrow \left\lfloor \frac{\lfloor mx \rfloor}{m} \right\rfloor = n$$

$$\left\lfloor \frac{\lfloor mx \rfloor}{m} \right\rfloor = \lfloor x \rfloor.$$

Case 2: $m < 0$, let $x = 0.5$, $m = -1$

$$\lfloor x \rfloor = \lfloor 0.5 \rfloor = 0$$

$$\left\lfloor \frac{\lfloor -1 \times 0.5 \rfloor}{-1} \right\rfloor = \left\lfloor \frac{\lfloor -0.5 \rfloor}{-1} \right\rfloor = \left\lfloor \frac{-1}{-1} \right\rfloor = 1$$

$\therefore$ doesn't hold!

$\therefore x \in \mathbb{R}$ and $m \in \mathbb{Z}^+$

$$\left\lfloor \frac{\lfloor mx \rfloor}{m} \right\rfloor = \lfloor x \rfloor$$